Transformata Z

1) Să se determine transformatele Z pentru următoarele semnale

a) $X=(\underbrace{e^n h(n+4)}_{x_n})_{n\in\mathbb{Z}}$

$$x_n=\begin{cases} e^n ; n+4\geq 0 \\ 0 ; n+4<0 \end{cases} = \begin{cases} e^n ; n\geq -4 \\ 0 ; n<-4 \end{cases}$$

$$L_x(z)=\sum_{n\in\mathbb{Z}} e^n h(n+4)\cdot z^{-n}=\sum_{n\geq -4} e^n z^{-n}$$

$$=e^{-4}z^4+e^{-3}z^3+e^{-2}z^2+e^{-1}z^1+\sum_{n\geq 0} e^n\cdot z^{-n}=$$

$$=\left(\frac{z}{e}\right)^4+\left(\frac{z}{e}\right)^3+\left(\frac{z}{e}\right)^2+\frac{z}{e}+\sum_{n\geq 0}\left(e\cdot z^{-1}\right)^n$$

$$=\left(\frac{z}{e}\right)^4+\left(\frac{z}{e}\right)^3+\left(\frac{z}{e}\right)^2+\frac{z}{e}+\frac{1}{1-e\cdot z^{-1}}$$

$$=\left(\frac{z}{e}\right)^4+\left(\frac{z}{e}\right)^3+\left(\frac{z}{e}\right)^2+\frac{z}{e}+\frac{z}{z-e}$$

$$\left|\frac{e}{z}\right|<1 \Rightarrow |e|<|z|$$

b) $X=(n^2\cdot 5^n\cdot h(n))_{n\in\mathbb{Z}}$

$$X \xrightarrow[\text{prop6}]{z} -zL_y'(z) \qquad y=n5^n h(n)$$

$$y \xrightarrow[\text{prop6}]{z} -zL_v'(z) \qquad v=5^n h(n)$$

$V \xrightarrow{z} \frac{z}{z-5}$; $\left|\frac{5}{z}\right|<1$

$a^n h(n) \xrightarrow{z} \frac{z}{z-a}$ pt. $\left|\frac{a}{z}\right|<1$

$$-zL_V'(z) = -z\cdot\left(\frac{z}{z-5}\right)' = -z\cdot\frac{z-5-z\cdot 1}{(z-5)^2} = \frac{5z}{(z-5)^2}$$

$$-zL_Y'(z) = -z\cdot\left(\frac{5z}{(z-5)^2}\right)' = -z\cdot\frac{5(z-5)^2 - 5z\cdot 2(z-5)}{(z-5)^4}$$

$$= -z\,\frac{5z-25-10z}{(z-5)^3} = \frac{5z^2+25z}{(z-5)^3}$$

$$L_X(z) = -zL_Y'(z) = \frac{5z^2+25z}{(z-5)^3}$$

c) $X = (e^{-an}\operatorname{sh} 2n\, h(n))_{n\in\mathbb{Z}}$ $\quad a\in\mathbb{C}$

$$X = \left((e^{-a})^n \underbrace{\frac{e^{2n}-e^{-2n}}{2} h(n)}_{Y}\right)_{n\in\mathbb{Z}}$$

$\downarrow z$ asem.

$$L_Y\left(\frac{z}{e^{-a}}\right) = L_Y(ze^a)$$

$$L_Y(z) = \frac{1}{2}L_{e^{2n}h(n)}(z) - \frac{1}{2}L_{e^{-2n}h(n)}(z)$$

$$= \frac{1}{2}L_{h(n)}\left(\frac{z}{e^2}\right) - \frac{1}{2}L_{h(n)}\left(\frac{z}{e^{-2}}\right)$$

$$L_{h(n)}(z) = \frac{z}{z-1}$$

$$L_y(z) = \frac{1}{2}\cdot\frac{\frac{z}{e^2}}{\frac{z}{e^2}-1} - \frac{1}{2}\frac{\frac{z}{e^{-2}}}{\frac{z}{e^{-2}}-1} = \frac{1}{2}\cdot\frac{z}{z-e^2} - \frac{1}{2}\frac{z}{z-e^{-2}}$$

$$L_y(z) = \frac{1}{2}\cdot\frac{z}{z-e^2} - \frac{1}{2}\frac{ze^2}{ze^2-1}$$

$$L_y(ze^a) = \frac{1}{2}\left(\frac{ze^a}{ze^a-e^2} - \frac{ze^{a+2}}{ze^{a+2}-1}\right)$$

d) $x = \left(e^{n-4}h(n-4)\right)_{n\in\mathbb{Z}}$

$$x = (e^n h(n)) * \delta_4 \xrightarrow{z} L[e^n h(n)](z)\cdot z^{-4} =$$

$$a^n h(n) \xrightarrow{z} \frac{z}{z-a} \quad pt.\ \left|\frac{a}{z}\right|<1$$

$$= \frac{z}{z-e}\cdot z^{-4} \quad pt.\ \left|\frac{e}{z}\right|<1$$

e) $x = \left((-1)^n \cdot n\, h(n)\right)_{n\in\mathbb{Z}}$ (x_n)

$$L_x(z) = \sum_{n\in\mathbb{Z}} (-1)^n h(n)\cdot z^{-n} = \sum_{n\geq 0} (-1)^n z^{-n} = \sum_{n\geq 0} (-1)^n (z^{-1})^n$$

$$= \frac{1}{1+\frac{1}{z}} = \frac{z}{z+1} \quad pt.\ \left|\frac{1}{z}\right|<1$$

$$(-1)^n \cdot n\, h(n) \xrightarrow{z} -z\cdot L_x'(z) = -z\cdot\left(\frac{z}{z+1}\right)'$$

$$= -z\cdot\frac{z+1-z}{(z+1)^2} = \frac{-z}{(z+1)^2}$$

② Să se determine șirul a cărui transformată Z este:

a) $L_x(z) = \frac{z}{(z-1)(z^2+1)}$

$$L_x(z) = \frac{z}{(z-1)(z-i)(z+i)}$$

$\{1; \pm i\}$ poli de ordinul 1

$$x_n = \text{Rez}\left(z^{n-1} \cdot L_x(z), 1\right) + \text{Rez}\left(z^{n-1} \cdot L_x(z), i\right) + \text{Rez}\left(z^{n-1} \cdot L_x(z), -i\right)$$

$$\text{Rez}\left(z^{n-1} \cdot L_x(z), 1\right) = \lim_{z \to 1} (z-1) \cdot \frac{z^n}{(z-1)(z-i)(z+i)} = \frac{1}{(1-i)(1+i)} = \frac{1}{2}$$

$$\text{Rez}\left(z^{n-1} \cdot L_x(z), i\right) = \lim_{z \to i} (z-i) \frac{z^n}{(z-1)(z-i)(z+i)} = \frac{i^n}{(i-1) \cdot 2i} = \frac{i^n}{-2(1+i)}$$

$$\text{Rez}\left(z^{n-1} \cdot L_x(z), -i\right) = \lim_{z \to -i} (z+i) \frac{z^n}{(z-1)(z-i)(z+i)} = \frac{(-i)^n}{-(1+i)(-2i)} = \frac{(-i)^n}{2(i-1)}$$

$$x_n = \frac{1}{2} - \frac{i^n}{2(1+i)} + \frac{(-i)^n}{2(i-1)}$$

$$x_{4k} = \frac{1}{2} - \frac{1}{2(1+i)} + \frac{1}{2(i-1)}$$

$$x_{4k+1} = \frac{1}{2} - \frac{i}{2(1+i)} + \frac{-i}{2(i-1)} = \frac{1}{2} + \frac{i}{2(1+i)} + \frac{-i}{(1+i)2i}$$

$$= \frac{1}{2} + \frac{-1+i}{2(1+i)}$$

$$x_{4k+2} = \frac{1}{2} - \frac{-1}{2(1+i)} + \frac{-1}{2(i-1)}$$

$$x_{4k+3} = \frac{1}{2} - \frac{-i}{2(1+i)} + \frac{i}{2(i-1)} = \frac{1}{2} + \frac{-i}{2(1+i)} + \frac{i}{(1+i)2i}$$

$$= \frac{1}{2} + \frac{1-i}{2(1+i)}$$

b) $L_x(z) = \frac{z}{(z-3)^2}$

$$x_k = \operatorname{Rez}(z^{k-1} L_x(z), 3) = \lim_{z \to 3} \left((z-3)^2 \cdot \frac{z^k}{(z-3)^2} \right)^{(1)}$$

$$x_k = \lim_{z \to 3} k z^{k-1} = k 3^{k-1} \qquad \forall k \in \mathbb{N}$$

③ Să se determine semnalul discret pozitiv, rezolvând $y * a = x$ pentru

a) $a = \delta_{-2} - 4\delta_{-1} + 3 \qquad y \in S_d^+$

$x = (2h(k))_{k \in \mathbb{Z}}$

$$L_y(z) \cdot L_a(z) = L_x(z) \Rightarrow L_y(z) = \frac{L_x(z)}{L_a(z)}$$

$$L_a(z) = z^2 - 4z + 3$$

$$L_x(z) = \sum_{k \in \mathbb{Z}} 2h(k) \cdot z^{-k} = 2 \sum_{k \geq 0} z^{-k} = 2 \cdot \frac{1}{1 - z^{-1}} = \frac{2z}{z-1}$$

$$\Rightarrow L_y(z) = \frac{2z}{(z^2 - 4z + 3)(z-1)}$$

$\Delta = 16 - 4\cdot 1\cdot 3 = 16 - 12 = 4 \Rightarrow z_{1,2} = \frac{4 \pm 2}{2} \begin{matrix} \nearrow 3 \\ \searrow 1 \end{matrix}$

$\{3\}$ pol de ordinul 1

$\{1\}$ pol de ordinul 2

$$y_n = \operatorname{Rez}(L_y(z) z^{n-1}, 1) + \operatorname{Rez}(L_y(z)\cdot z^{n-1}, 3) \quad \forall n \in \mathbb{N}$$

$$\operatorname{Rez}(L_y(z) z^{n-1}, 1) = \lim_{z\to 1} \left((z-1)^2 \cdot \frac{2z^n}{(z-1)^2(z-3)} \right)'$$

$$= \lim_{z\to 1} \frac{2nz^{n-1}(z-3) - 2z^n}{(z-3)^2} = \lim_{z\to 1} \frac{z^n(2n-2) - 6nz^{n-1}}{(z-3)^2}$$

$$= \frac{2n-2-6n}{(-2)^2} = \frac{-4n-2}{4} = -n - \frac{1}{2}$$

$$\operatorname{Rez}(L_y(z)\cdot z^{n-1}, 3) = \lim_{z\to 3} (z-3)\cdot \frac{2z^n}{(z-1)^2(z-3)} = \frac{2\cdot 3^n}{4}$$

$$y_n = -n - \frac{1}{2} + \frac{1}{2} 3^n \quad \forall n \in \mathbb{N}$$

b) $a = \delta_{-1} - 2\delta$ $\qquad y \in S_d^+$

$$x = \left((n^2 - 2n - 1) h(n) \right)_{n\in\mathbb{Z}}$$

$$L_y(z)\cdot L_a(z) = L_x(z) \Rightarrow L_y(z) = \frac{L_x(z)}{L_a(z)}$$

$$L_a(z) = z - 2$$

$$h(n) \xrightarrow{z} \frac{z}{z-1}$$

$$nh(n) \xrightarrow{z} -z\cdot \left(\frac{z}{z-1} \right)' = -z\, \frac{z-1-z}{(z-1)^2} = \frac{z}{(z-1)^2}$$

$$u^2 h(u) \xrightarrow{\mathcal{Z}} -z\cdot\left(\mathcal{L}[uh(u)](z)\right)' = -z\cdot\frac{(z-1)^2 - z\cdot 2(z-1)}{(z-1)^4 3}$$

$$= -z\cdot\frac{z-1-2z}{(z-1)^3} = \frac{z(z+1)}{(z-1)^3}$$

$$L_x(z) = \frac{z(z+1)}{(z-1)^3} - 2\cdot\frac{z}{(z-1)^2} - \frac{z}{z-1}$$

$$= \frac{z(z+1) - 2z(z-1) - z(z-1)^2}{(z-1)^3} = \frac{z^2+z-2z^2+2z-z(z-1)^2}{(z-1)^3}$$

$$L_x(z) = \frac{-z^2+3z-z(z-1)^2}{(z-1)^3} = \frac{z(-z+3-z^2+2z-1)}{(z-1)^3}$$

$$L_x(z) = \frac{z(-z^2+z+2)}{(z-1)^3}$$

$$L_y(z) = \frac{z(-z^2+z+2)}{(z-1)^3(z-2)} = \frac{z(z-2)(z+1)}{(z-1)^3(z-2)} = \frac{z(z+1)}{(z-1)^3}$$

$$\Delta = 1+4\cdot 1\cdot 2 = 9 \qquad z_{1,2} = \frac{-1\pm 3}{-2} \longrightarrow \frac{-4}{-2} = 2 \quad \searrow \frac{2}{-2} = -1$$

$z=1$ pol de ordinul 3

$$y_u = \operatorname{Rez}\left(L_y(z)\, z^{u-1}, 1\right) \qquad \forall u \in \mathbb{N}$$

$$y_u = \lim_{z\to 1}\frac{1}{2!}\left((z-1)^3\cdot\frac{z(z+1)}{(z-1)^3}\cdot z^{u-1}\right)^{(2)} = \frac{1}{2}\lim_{z\to 1}\left(z^u(z+1)\right)^{(2)}$$

$$= \frac{1}{2}\lim_{z\to 1}\left(z^{u+1}+z^u\right)^{(2)} = \frac{1}{2}\lim_{z\to 1}\left((u+1)z^u + uz^{u-1}\right)'$$

$$= \frac{1}{2}\lim_{z\to 1}\left(u(u+1)z^{u-1} + u(u-1)z^u\right) = \frac{1}{2}u\left(u+1+u-1\right)$$

$$= \frac{1}{2}u\cdot 2u = u^2$$

4) Să se afle șirurile $(x_n)_{n\in\mathbb{N}}$ date prin relația de recurență:

a) $x_0 = 1$

$x_1 = 3$

$x_{n+2} - 7x_{n+1} + 12x_n = 1 \quad \forall n \in \mathbb{N}$

Considerăm $(x_n)_{n\in\mathbb{N}}$ restricția la $\mathbb{N}$ a unui șir cu suport pozitiv

$x_{n+2} - 7x_{n+1} + 12x_n = y_n \quad \forall n \in \mathbb{Z}$

Pt. $n=-2 \Rightarrow \underbrace{x_0}_{1} - 7\underbrace{x_{-1}}_{0} + 12\underbrace{x_{-2}}_{0} = y_{-2} \Rightarrow y_{-2} = 1 \Rightarrow$

Pt. $n=-1 \Rightarrow \underbrace{x_1}_{3} - 7\underbrace{x_0}_{1} + 12\underbrace{x_{-1}}_{0} = y_{-1} \Rightarrow y_{-1} = 3-7$

$y_{-1} = -4$

$$\Rightarrow y_n = \begin{cases} 1 & n \geq 0 \\ -4 & n = -1 \\ 1 & n = -2 \\ 0 & n \leq -3 \end{cases} \qquad y = h_n - 4\delta_{-1} + \delta_{-2} \quad \forall n \in \mathbb{Z}$$

$$(x * \delta_{-2})(n) - 7(x * \delta_{-1})(n) + 12(x * \delta)(n) = y_n$$

$$x * \underbrace{(\delta_{-2} - 7\delta_{-1} + 12\delta)}_{a} = y \Rightarrow L_x(z) \cdot L_a(z) = L_y(z)$$

$$\Rightarrow L_x(z) = \frac{L_y(z)}{L_a(z)}$$

$$L_a(z) = z^2 - 7z + 12$$

$$L_y(z) = \frac{z}{z-1} - 4z + z^2 = \frac{z - 4z^2 + 4z + z^3 - z^2}{z-1}$$

$$= \frac{z^3 - 5z^2 + 5z}{z-1} \Longrightarrow$$

$$\Longrightarrow L_x(z) = \frac{z^3 - 5z^2 + 5z}{(z-1)(z^2 - 7z + 12)}$$

$z_1 = 1$

$\Delta = 49 - 4 \cdot 1 \cdot 12 = 49 - 48 = 1 \Rightarrow z_{2,3} = \frac{7 \pm 1}{2} \begin{cases} \nearrow 4 \\ \searrow 3 \end{cases}$

$z_{1,2,3}$ poli de ordinul 1

$$x_u = \text{Rez}(L_x(z) \cdot z^{u-1}, 1) + \text{Rez}(L_x(z) \cdot z^{u-1}, 3) + \text{Rez}(L_x(z) \cdot z^{u-1}, 4)$$

$$\text{Rez}(L_x(z) \cdot z^{u-1}, 1) = \lim_{z \to 1} (z-1) \cdot \frac{z^3 - 5z^2 + 5z}{(z-1)(z^2 - 7z + 12)} \cdot z^{u-1}$$

$$= \frac{1 - 5 + 5}{1 - 7 + 12} = \frac{1}{6}$$

$$\text{Rez}(L_x(z) \cdot z^{u-1}, 3) = \lim_{z \to 3} (z-3) \frac{z^3 - 5z^2 + 5z}{(z-1)(z^2 - 7z + 12)} \cdot z^{u-1}$$

$z - 4$

$$= \frac{27 + 5 \cdot 3(-3+1)}{2 \cdot (-1)} \cdot 3^{u-1} = \frac{27 + 15 \cdot (-2)}{-2} \cdot 3^{u-1} = \frac{3}{2} \cdot 3^{u-1} = \frac{3^u}{2}$$

$$\text{Rez}(L_x(z) \cdot z^{u-1}, 4) = \lim_{z \to 4} (z-4) \cdot \frac{z^3 - 5z^2 + 5z}{(z-1)(z^2 - 7z + 12)} \cdot z^{u-1}$$

$z - 3$

$$= \frac{4(16-20+5)}{3\cdot 1}\cdot 4^{n-1} = \frac{4^n}{3}$$

$$x_n = \frac{1}{6} + \frac{3^n}{2} + \frac{4^n}{3}$$

b) $x_0 = 1$

$x_1 = -1$

$x_{n+2} - 7x_{n+1} + 10x_n = 3n \quad \forall n \in \mathbb{N}$

Considerăm $(x_n)_{n\in\mathbb{N}}$ restricția la $\mathbb{N}$ a unui semnal cu suport pozitiv

$x_{n+2} - 7x_{n+1} + 10x_n = y_n \quad \forall n \in \mathbb{Z}$

Pt. $n=-2 \Rightarrow \underbrace{x_0}_{1} - 7\underbrace{x_{-1}}_{0} + 10\underbrace{x_{-2}}_{0} = y_{-2} \Rightarrow y_{-2} = 1$

Pt. $n=-1 \Rightarrow \underbrace{x_1}_{-1} - 7\underbrace{x_0}_{1} + 10\underbrace{x_{-1}}_{0} = y_{-1} \Rightarrow y_{-1} = -8$

$$\Rightarrow y_n = \begin{cases} 3n & n \geq 0 \\ -8 & n = -1 \\ 1 & n = -2 \\ 0 & n \leq -3 \end{cases} \Rightarrow y = 3n\,h(n) - 8\delta_{-1} + \delta_{-2}$$

$\forall n \in \mathbb{Z}$

$(x * \delta_{-2})(n) - 7(x * \delta_{-1})(n) + 10(x * \delta)(n) = y_n$

$x * (\underbrace{\delta_{-2} - 7\delta_{-1} + 10\delta}_{a}) = y$

$L_x(z)\cdot L_a(z) = L_y(z) \Rightarrow L_x(z) = \frac{L_y(z)}{L_a(z)}$

$L_a(z) = z^2 - 7z + 10 = (z-2)(z-5)$

$$L_y(z) = \frac{3z}{(z-1)^2} - 8z + z^2 = \frac{3z + (z^2-8z)(z-1)^2}{(z-1)^2}$$

$$L_y(z) = \frac{z[3+(z-8)(z^2-2z+1)]}{(z-1)^2} = \frac{z[z^3-10z^2+17z-5]}{(z-1)^2}$$

$$L_x(z) = \frac{z[z^3-10z^2+17z-5]}{(z-1)^2(z-2)(z-5)}$$

$z=1$ pol de ordinul 2

$z \in \{2; 5\}$ poli de ordinul 1

$$x_u = \text{Rez}\left(L_x(z)\cdot z^{u-1}, 1\right) + \text{Rez}\left(L_x(z)\cdot z^{u-1}, 2\right) + \text{Rez}\left(L_x(z)\cdot z^{u-1}, 5\right)$$

$$\text{Rez}\left(L_x(z)\cdot z^{u-1}, 1\right) = \lim_{z\to 1}\left((z-1)^2 \frac{z^u[z^3-10z^2+17z-5]}{(z-1)^2(z-2)(z-5)}\right)'$$

$$= \lim_{z\to 1} \frac{[uz^{u-1}(z^3-10z^2+17z-5) + z^u(3z^2-20z+17)](z-2)(z-5)}{(z-2)^2(z-5)^2}$$

$$\frac{-z^u(z^3-10z^2+17z-5)\cdot[z-5+z-2]}{}$$

$$= \lim_{z\to 1} \frac{[u(1-10+17-5)+(3-20+17)](-1)(-4) - (1-10+17-5)(-5)}{(-1)^2\cdot(-4)^2}$$

$$= \frac{[u\cdot 3+0)4+5\cdot 3}{(-1)^2(-4)^2} = \frac{3\cdot 4u+15}{16} = \frac{12u+15}{16}$$

$$\text{Rez}\left(L_x(z)\cdot z^{u-1}, 2\right) = \lim_{z\to 2}(z-2)\frac{z^u[z^3-10z^2+17z-5]}{(z-1)^2(z-2)(z-5)} =$$

$$= \frac{2^n(8-40+34-5)}{1^2\cdot(-3)} = \frac{-2^n\cdot(-3)}{3} = 2^n$$

$$\operatorname{Rez}\left(L_x(z)\cdot z^{n-1}, 5\right) = \lim_{z\to 5}(z-5)\,\frac{z^n[z^3-10z^2+17z-5]}{(z-1)^2(z-2)(z-5)} =$$

$$= \frac{5^n(125-250+85-5)}{4^2\cdot 3} = \frac{5^n\cdot(-45)}{16\cdot 3} = \frac{-3\cdot 5^{n+1}}{16}$$

$$X_n = \frac{12n+15}{16} + 2^n - \frac{3\cdot 5^{n+1}}{16}$$

c) $X_0 = 0$

$X_1 = -1$

$X_{n+2} + X_{n+1} - 6X_n = n\cdot 2^n \quad \forall n\in\mathbb{N}$

Considerăm $(X_n)_{n\in\mathbb{N}}$ restricția la $\mathbb{N}$ a unui semnal cu suport pozitiv

$X_{n+2} + X_{n+1} - 6X_n = y_n \quad \forall n\in\mathbb{Z}$

Pt. $n=-2 \Rightarrow X_0 + X_{-1} - 6X_{-2} = y_{-2} \Rightarrow y_{-2} = 0$

(0 = 0, 0 = 0, 0 = 0)

Pt. $n=-1 \Rightarrow X_1 + X_0 - 6X_{-1} = y_{-1} \Rightarrow y_{-1} = -1$

(−1 = −1, 0 = 0, 0 = 0)

$$y_n = \begin{cases} n2^n & ; \forall n\geq 0 \\ -1 & ; n=-1 \\ 0 & ; n\leq -2 \end{cases} \Rightarrow y(n) = n2^n h(n) - \delta_{-1} \quad \forall n\in\mathbb{Z}$$

$$(x*\delta_{-2})(n)+(x*\delta_{-1})(n)-6(x*\delta)(n)=y_n$$

$$x*\underbrace{(\delta_{-2}+\delta_{-1}-6\delta)}_{a}=y$$

$$L_x(z)\cdot L_a(z)=L_y(z)\Rightarrow L_x(z)=\frac{L_y(z)}{L_a(z)}$$

$$L_a(z)=z^2+z-6=(z-2)(z+3)$$

$$n2^n h(n)\xrightarrow{z}-zL_{2^n}'(z)$$

$$\Rightarrow$$

$$L_{2^n h(n)}(z)=L_{h(n)}\left(\frac{z}{2}\right)=\frac{1}{1-\frac{2}{z}}=\frac{z}{z-2}$$

$$L_y(z)=-z\cdot\left(\frac{z}{z-2}\right)'-8z=-z\cdot\frac{z-2-z}{(z-2)^2}-8z$$

$$=\frac{2z}{(z-2)^2}-z=\frac{2z-z(z-2)^2}{(z-2)^2}=\frac{2z-z(z^2-4z+4)}{(z-2)^2}$$

$$=\frac{2z-z^3+4z^2-4z}{(z-2)^2}=\frac{-z^3+4z^2-2z}{(z-2)^2}=(-z)\frac{z^2-4z+2}{(z-2)^2}$$

$$L_x(z)=\frac{L_y(z)}{L_a(z)}=\frac{-z(z^2-4z+2)}{(z-2)^2(z-2)(z+3)}=\frac{-z(z^2-4z+2)}{(z-2)^3(z+3)}$$

$z=2$ pol de ordinul 3

$z=-3$ pol de ordinul 1

$$x_n=\operatorname{Rez}\left(L_x(z)\cdot z^{n-1},1\right)+\operatorname{Rez}\left(L_x(z)\cdot z^{n-1},3\right)$$

$$x_n=\frac{1}{2}\lim_{z\to 2}\left(\frac{-z^n(z^2-4z+2)}{z+3}\right)^{(2)}+\lim_{z\to 3}\left(\frac{-z^n(z^2-4z+2)}{(z-2)^3}\right)$$

$\forall n\in\mathbb{N}$